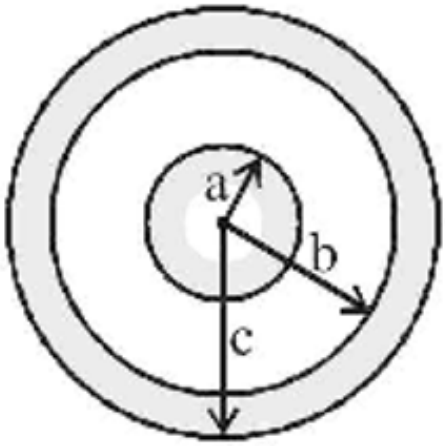
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A solid conducting sphere of radius a has a net positive charge $2Q$. A conducting spherical shell of inner radius b and outer radius c is concentric with the solid sphere and has a net charge - Q.
The surface charge density on the inner and outer surfaces of the spherical shell will be respectively



(A) $-\frac{2Q}{4\pi b^2}, \frac{Q}{4\pi c^2}$

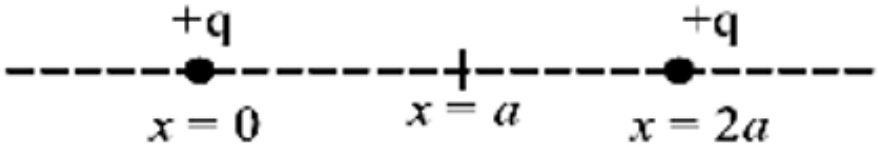
(B) $-\frac{Q}{4\pi b^2}, \frac{Q}{4\pi c^2}$

(C) $0, \frac{Q}{4\pi c^2}$

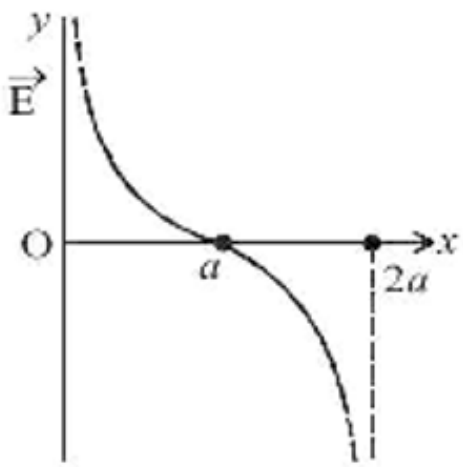
(D) $\frac{Q}{4\pi c^2}, 0$

Q.2

Figure shows two charges of equal magnitude separated by a distance $2a$. As we move away from the charge situated at $x = 0$ to the charge situated at $x = 2a$, which of the following graphs shows the correct behaviour of electric field ?

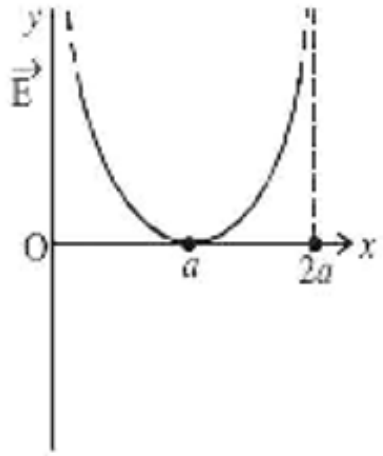


(A)

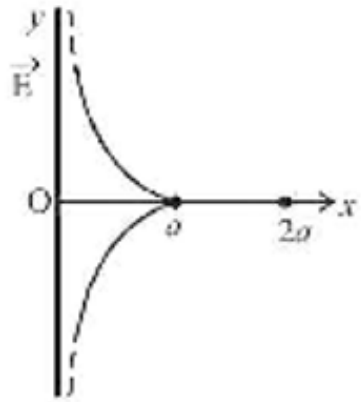


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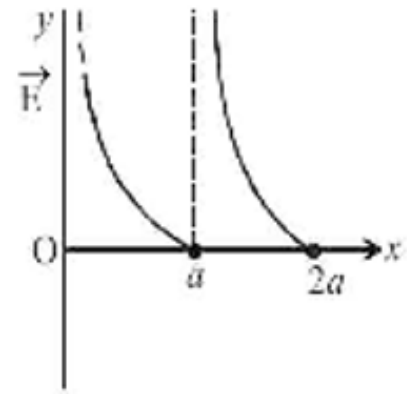
(B)



(C)

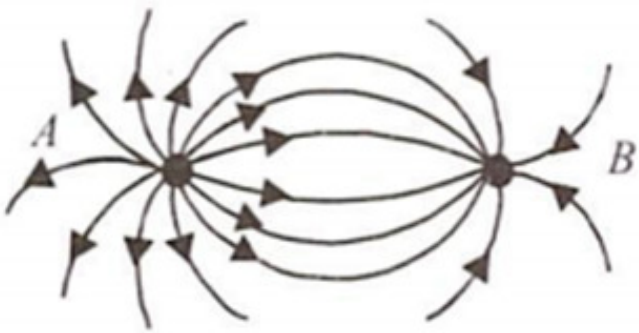


(D)



Q.3

The spatial distribution of electric field due to charges (A, B) is shown in figure. Which one of the following statements is correct?



(A) A is $+\mathrm{ve}$ and $\mathrm{B}-\mathrm{ve}, |\mathrm{A}| > |\mathrm{B}|$

(B) A is $-\mathrm{ve}$ and $\mathrm{B}+\mathrm{ve}, |\mathrm{A}| = |\mathrm{B}|$

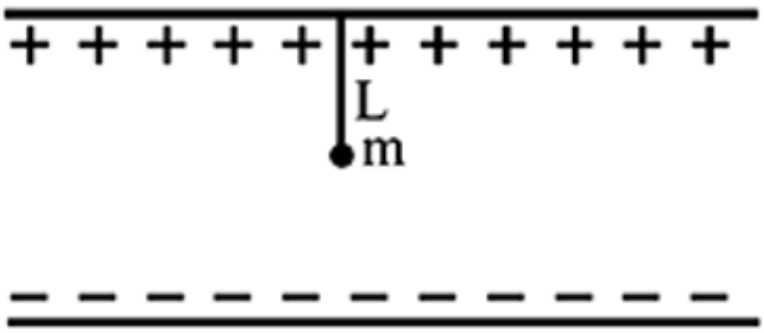
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(C) Both are +ve but $A > B$

(D) Both are $-$ ve but $A > B$

Q.4

A small sphere carrying a charge ' q ' is hanging in between two parallel plates by a string of length L. Time period of pendulum is T_0. When parallel plates are charged, the time period changes to T. The ratio T/T_0 is equal to



(A) $\left(\frac{g+\frac{qE}{m}}{g}\right)^{1/2}$

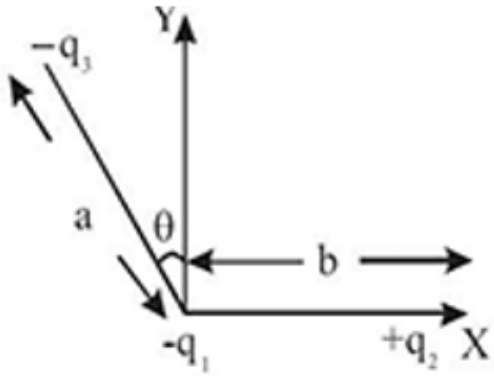
(B) $\left(\frac{g}{g+\frac{qE}{m}}\right)^{3/2}$

(C) $\left(\frac{g}{g+\frac{qE}{m}}\right)^{1/2}$

(D) None of these

Q.5

Three charges $-q_1, +q_2$ and $-q_3$ are placed as shown in the figure. The x = component of the force on $-q_1$ is proportional to



(A) $\frac{q_2}{b^2} - \frac{q_3}{a^2}\cos\theta$

(B) $\frac{q_2}{b^2} + \frac{q_3}{a^2}\sin\theta$

(C) $\frac{q_2}{b^2} + \frac{q_3}{a^2}\cos\theta$

(D) $\frac{q_2}{b^2} - \frac{q_3}{a^2}\sin\theta$

Q.6

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An oil drop of radius r and density ρ is held stationary in a uniform vertically upwards electric field 'E'. If $\rho_0(<\rho)$ is the density of air and e is quanta of charge, then the drop has-

(A) $\frac{4\pi r^3(\rho-\rho_0)g}{3\mathrm{eE}}$ excess electrons

(B) $\frac{4\pi r^2(\rho-\rho_0)g}{\mathrm{eE}}$ excess electrons

(C) deficiency of $\frac{4\pi r^3(\rho-\rho_0)g}{3\mathrm{eE}}$ electrons

(D) deficiency of $\frac{4\pi r^2(\rho-\rho_0)g}{\mathrm{eE}}$ electrons

Q.7

Which of the following is a wrong statement?

(A) The charge of an isolated system is conserved

(B) It is not possible to create or destroy charged particles

(C) It is possible to create or destroy charged particles

(D) It is not possible to create or destroy net charge

Q.8

Two positive ions, each carrying a charge q, are separated by a distance d. If F is the force of repulsion between the ions, the number of electrons missing from each ion will be (e being the charge of an electron)

(A) $\frac{4\pi\varepsilon_0 F d^2}{e^2}$

(B) $\sqrt{\frac{4\pi\varepsilon_0 F e^2}{d^2}}$

(C) $\sqrt{\frac{4\pi\varepsilon_0 F d^2}{e^2}}$

(D) $\frac{4\pi\varepsilon_0 F d^2}{q^2}$

Q.9

Two small similar metal spheres A and B having charges $4\mathrm{q}$ and $-4\mathrm{q}$, when placed at a certain distance apart, exert an electric force F on each other. When another identical uncharged sphere C, first touched with A then with B and then removed to infinity, the force of interaction between A and B for the same separation will be $\frac{F}{x}$, then find the value of x?

(A) 7

(B) 8

(C) 9

(D) 10

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Q.10

Two balls of same mass and carrying equal charge are hung from a fixed support of length l. At electrostatic equilibrium, assuming that angles made by each thread is small, the separation, x between the balls is proportional to :

(A) l

(B) l^2

(C) $l^{2/3}$

(D) $l^{1/3}$

Q.11

The force of repulsion between two electrons at a certain distance is F. The force between two protons separated by the same distance is $\left(\mathrm{m_p} = 1836\ \mathrm{m_e}\right)$

(A) $2\ \mathrm{F}$

(B) F

(C) $1836\ \mathrm{F}$

(D) $\frac{\mathrm{F}}{1836}$

Q.12

The force between two small charged spheres having charges of $1 \times 10^{-7}\mathrm{C}$ and $2 \times 10^{-7}\mathrm{C}$ placed $20\ \mathrm{cm}$ apart in air is

(A) $4.5 \times 10^{-2}\ \mathrm{N}$

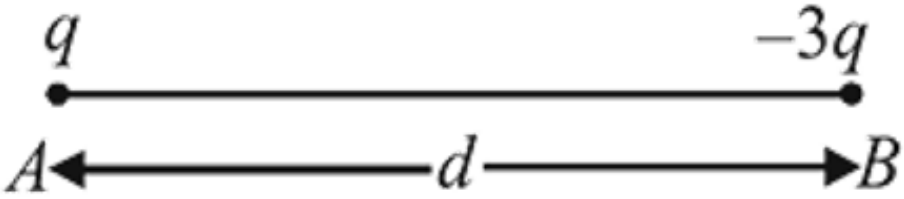
(B) $4.5 \times 10^{-3}\ \mathrm{N}$

(C) $5.4 \times 10^{-2}\ \mathrm{N}$

(D) $5.4 \times 10^{-3}\ \mathrm{N}$

Q.13

Two charge q and $-3q$ are placed fixed on x -axis separated by distance d. Where should a third charge $2q$ be placed such that it will not experience any force?



(A) $\frac{d-\sqrt{2}d}{2}$

(B) $\frac{d+\sqrt{3}d}{2}$

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(C) $\frac{d+3d}{2}$

(D) $\frac{d-\sqrt{5}d}{2}$

Q.14

Two insulated charged metalic sphere P and Q have their centres separated by a distance of $60\ \mathrm{cm}$. The radii of P and Q are negligible compared to the distance of separation. The mutual force of electrostatic repulsion if the charge on each is $3.2 \times 10^{-7}\mathrm{C}$ is

(A) $5.2 \times 10^{-4} N$

(B) $2.5 \times 10^{-3} N$

(C) $1.5 \times 10^{-3} N$

(D) $3.5 \times 10^{-4} N$

Q.15

Four charges each equal to Q are placed at the four corners of a square and a charge q is placed at the centre of the square. If the system is in equilibrium, the value of q is:

(A) $\frac{Q}{2}(1+2\sqrt{2})$

(B) $\frac{-Q}{4}(1+2\sqrt{2})$

(C) $\frac{Q}{4}(1+2\sqrt{2})$

(D) $\frac{-Q}{2}(1+2\sqrt{2})$

Q.16

Two unit negative charges are placed on a straight line. A positive charge q is placed exactly at the midpoint between these unit charges. If the system of these three, charges is in equilibrium, the value of $\mathrm{q}(\text{ in } \mathrm{C})$ is:

(A) 1.0

(B) 0.75

(C) 0.5

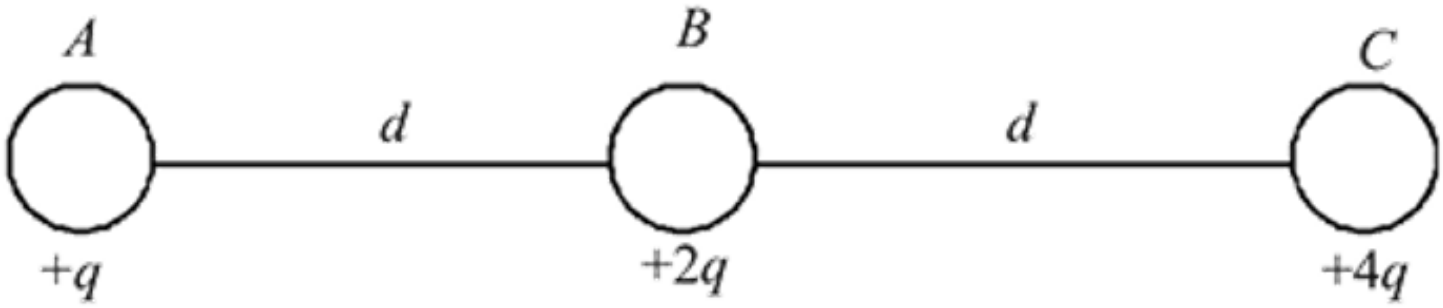
(D) 0.25

Q.17

Three charges $+q, +2q$ and $+4q$ are connected by strings as shown in the figure. What is ratio of tensions in

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the strings AB and BC ?



(A) $1:2$

(B) $1:3$

(C) $2:1$

(D) $3:1$

Q.18

$1\mathrm{C}$ charge is equivalent to charge on how much number of protons?

(A) 6×10^{18}

(B) 7×10^{19}

(C) 8×10^{20}

(D) 9×10^{21}

Q.19

The metal knob of a gold leaf electroscope is touched with a positively charged rod. When it is taken away the leaves stay separated. Now the metal knob is touched by negatively charged rod. The separation between the leaves

(A) increases

(B) decreases

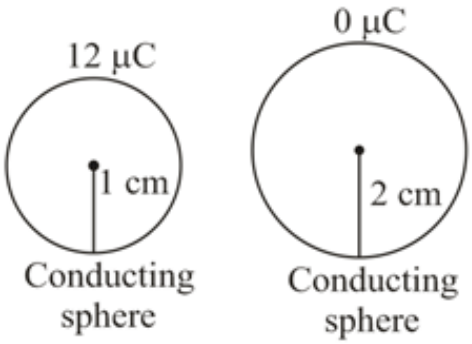
(C) remains same

(D) first increases then decreases.

Q.20

When these two spheres are made to touch with each other, then final charge on bigger sphere will be:

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(A) $0\ \mu C$

(B) $12\ \mu C$

(C) $4\ \mu C$

(D) $8\ \mu C$

Answers & Solutions

Q.1 Answer:

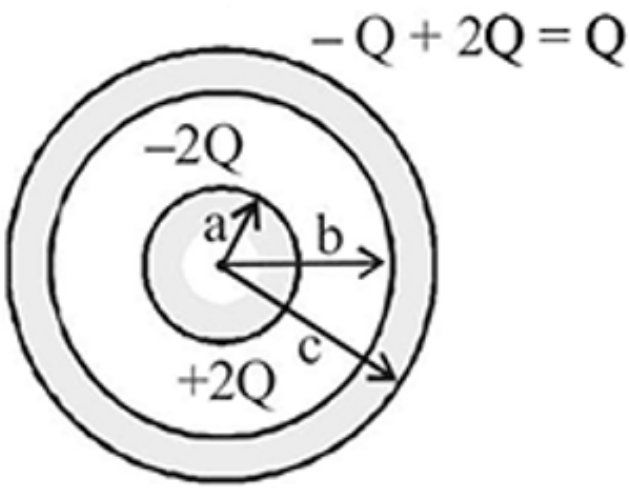
$-\frac{2Q}{4\pi b^2}, \frac{Q}{4\pi c^2}$

Solution:

$$\text{Surface charge density } (\sigma) = \frac{\text{Charge}}{\text{Surface area}}$$

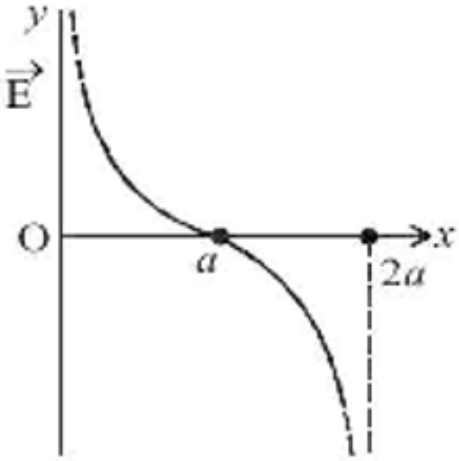
So $\sigma_{\text{inner}} = \frac{-2Q}{4\pi b^2}$

and $\sigma_{\text{Outer}} = \frac{Q}{4\pi c^2}$



Q.2 Answer:

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Solution:

For the distances close to the charge at $x = 0$ the field is very high and is in positive direction of x - axis. As we move towards the other charge the net electric field becomes zero at $\mathrm{x} = \mathrm{a}$ thereafter the influence of charge at $\mathrm{x} = 2$ a dominates and net field increases in negative direction of x-axis and grows unboundedly as we come closer and closer to the charge at $\mathrm{x} = 2\mathrm{a}$.

Q.3 Answer:

A is $+\mathrm{ve}$ and $\mathrm{B} - \mathrm{ve}, |\mathrm{A}| > |\mathrm{B}|$

Solution:

The lines proceed from the positive and lines will reach B, if B is negative.
Greater intensity of lines near the charge signifies greater magnitude of charge.
So, A is positive, B is negative and $|\mathrm{A}| > |\mathrm{B}|$.

Q.4 Answer:

$\left(\frac{g}{g+\frac{qE}{m}}\right)^{1/2}$

Solution:

$\mathrm{T}_0 = 2\pi\sqrt{\frac{\mathrm{L}}{\mathrm{g}}}$

When the plates are charged, the net acceleration is, $g' = g + a$

$\mathrm{g}' = \mathrm{g} + \frac{\mathrm{qE}}{\mathrm{m}} \quad \left(\mathrm{a} = \frac{\mathrm{qE}}{\mathrm{m}}\right)$

$\therefore \quad \mathrm{T} = 2\pi\sqrt{\frac{\mathrm{L}}{\mathrm{g}+\frac{\mathrm{qE}}{\mathrm{m}}}}$

$\therefore \quad \frac{T}{T_0} = \left(\frac{g}{g+\frac{qE}{m}}\right)^{1/2}$

Q.5 Answer:

$\frac{q_2}{b^2} + \frac{q_3}{a^2}\sin\theta$

Solution:

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Force on charge q_1 due to q_2 is

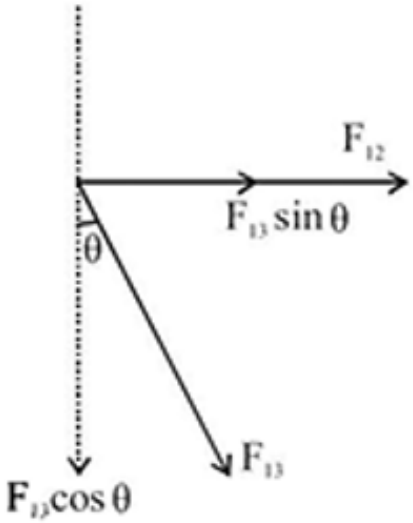
$$F_{12} = k\frac{q_1 q_2}{b^2}$$

Force on charge q_1 due to q_3 is

$$F_{13} = k\frac{q_1 q_3}{a^2}$$

The X - component of the force (F_x) on

$$\begin{aligned} q_1 \text{ is } & F_{12} + F_{13}\sin\theta \\ \therefore F_x &= k\tfrac{q_1 q_2}{b^2} + k\tfrac{q_1 q_2}{a^2}\sin\theta \\ \therefore F_x &\propto \tfrac{q_2}{b^2} + \tfrac{q_3}{a^2}\sin\theta \end{aligned}$$



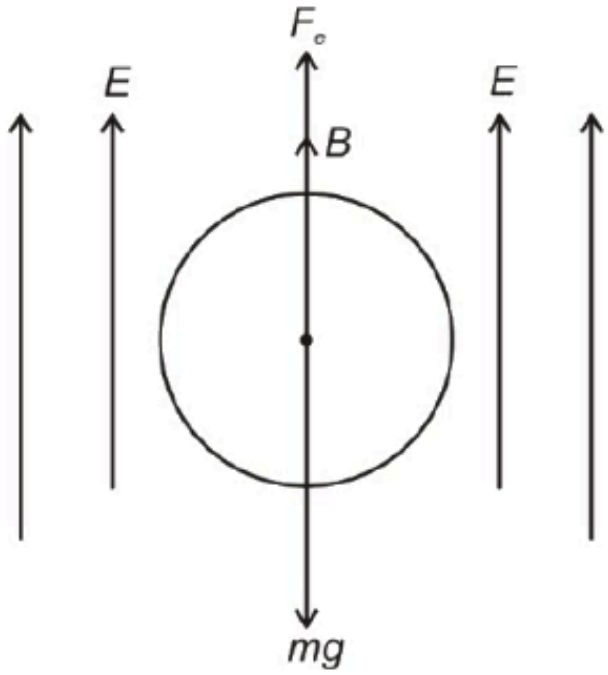
Q.6 Answer:

deficiency of $\frac{4\pi r^3(\rho-\rho_0)g}{3eE}$ electrons

Solution:

Net downward force on the drop $= \frac{4}{3}\pi r^3(\rho-\rho_0)g$ For equilibrium, electric force must be upwards i.e. charge on the drop is positive. $neE = \frac{4}{3}\pi r^3(\rho-\rho_0)g$ **i.e.** $n = \frac{4\pi r^3(\rho-\rho_0)g}{3eE}$

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For equilibrium of drop

Electric forces Buoyancy force

$F_e + B = mg$

Q.7 Answer:

It is not possible to create or destroy charged particles

Solution:

It is possible to create or destroy charged particles but it is not possible to create or destroy net charge. The charge of an isolated system is conserved.

Q.8 Answer:

$\sqrt{\frac{4\pi\varepsilon_0 F d^2}{e^2}}$

Solution:

Let n be the number of electrons missing. $F = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q^2}{d^2} \Rightarrow q = \sqrt{4\pi\varepsilon_0 d^2 F} = ne$

$\therefore \quad n = \sqrt{\frac{4\pi\varepsilon_0 F d^2}{e^2}}$

Q.9 Answer:

8

Solution:

$\mathrm{F} = \frac{1}{4\pi\varepsilon_0}\frac{(4q)(-4q)}{r^2}$

when C **is touched with** A**, then charge on** $\mathrm{A}\&\mathrm{C}$ **each** $= 2q$ **after that** C **is touched with** B**, charge on** $\mathrm{B} = \frac{2q+(-4q)}{2} = -q$

Now, force $\mathrm{F}' = \frac{1}{4\pi\varepsilon_0}\frac{(2q)(-q)}{r^2} \Rightarrow \mathrm{F}' = \frac{F}{8}$

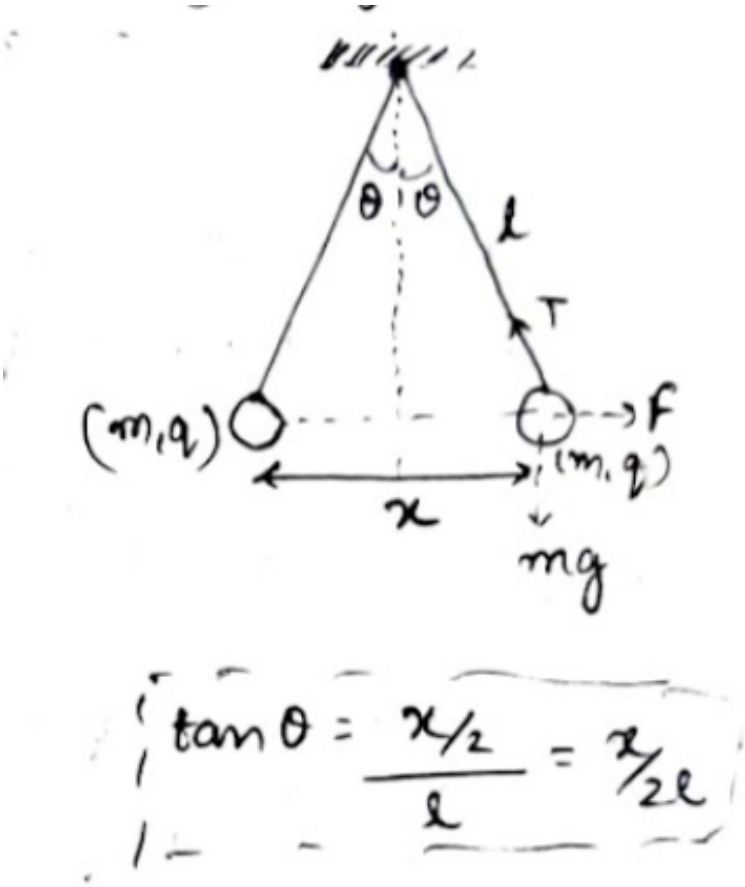
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Q.10 Answer:

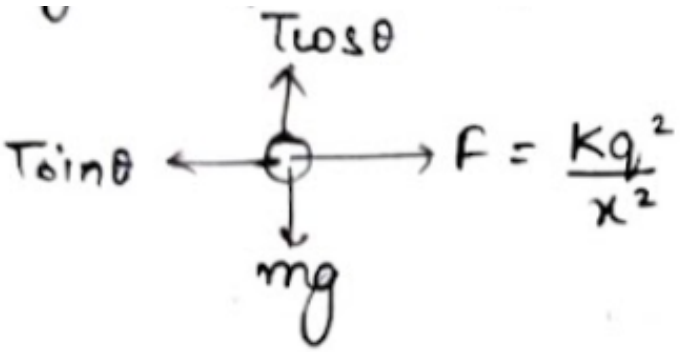
$l^{1/3}$

Solution:

The given situation can be represented by following diagram.



Free body diagram of a ball:



Balancing the forces we get:

$T \sin\theta = kq^2/x^2$ ... **(I)**

$T \cos\theta = mg$ ... **(II)**

Dividing (I) by (II)

$\tan\theta = \frac{kq^2}{mgx^2}$

$\Rightarrow \frac{x}{2l} = \frac{kq^2}{mgx^2}$

$\Rightarrow x^3 = \frac{2kq^2l}{mg}$

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$\Rightarrow x = \left(\frac{2kq^2}{mg}\right)^{1/3} l^{1/3}$

Hence, separation is proportional to $l^{1/3}$

Q.11 Answer:

F

Solution:

Electrostatic force is given by $\mathrm{F} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{\mathrm{r}^2}$
Here, charge and distance are same. So, force between two protons will be same

Q.12 Answer:

4.5×10^{-3} N

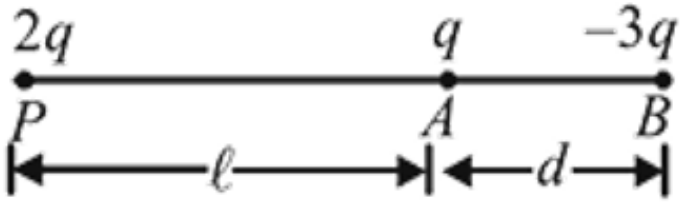
Solution:

Here, $q_1 = 1 \times 10^{-7}\mathrm{C}, q_2$ **and** $2 \times 10^{-1}\mathrm{C},$
$r = 20$ cm $= 20 \times 10^{-2}$ m
$F = \frac{q_1 q_2}{4\pi\varepsilon_0 r^2} = \frac{9\times10^9\times1\times10^{-7}\times2\times10^{-7}}{(20\times10^{-2})^2}$
$= 4.5 \times 10^{-3} N$

Q.13 Answer:

$\frac{d+\sqrt{3}d}{2}$

Solution:



Let a charge $2q$ **be placed at** P**, at a distance** I **from** A **where charge** q **is placed, as shown in figu color The charge** $2q$ **will not experience any force, when force, when force of repulsion on it due to** q **is balanced by force of attraction on it due to** $-3q$ **at** B **where** $\mathrm{AB} = d$
or $\frac{(2q)(q)}{4\pi\varepsilon_0\ell^2} = \frac{(2q)(-3q)}{4\pi\varepsilon_0(\ell+d)^2}$
$(\ell + d)^2 = 3\ell^2$ **or** $2\ell^2 - 2\ell d - d^2 = 0$
$\therefore \quad \ell = \frac{2d \pm \sqrt{4d^2+2d^2}}{4} = \frac{d}{2} \pm \frac{\sqrt{3}d}{2}$

Q.14 Answer:

$2.5 \times 10^{-3} N$

Solution:

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Here $q_1 = q_2 = 3.2 \times 10^{-7}\text{C}, r = 60\text{ cm}$
$= 0.6\text{ m}$
Electrostatic force, $F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$
$= \frac{9\times10^9\left(3.2\times10^{-7}\right)^2}{(0.6)^2} = 2.56 \times 10^{-3}\text{ N}$

Q.15 Answer:

$\frac{-Q}{4}(1+2\sqrt{2})$

Solution:

Step 1: Calculate all the required distances between the charges :

Let the side length of the square be a.

Refer Figure, In $\triangle\text{ACD}$,

$\Rightarrow \text{AD}^2 + \text{CD}^2 = \text{AC}^2$

$\Rightarrow a^2 + a^2 = AC^2 \quad \Rightarrow AC = \sqrt{2}a$

However, $\text{OC} = \frac{\text{AC}}{2} = \frac{\text{a}}{\sqrt{2}}$

Step 2 : Equilibrium of central charge

By symmetry, Force on central charge will be equal and opposite due to the diagonally opposite charges, which will cancel each other.

Hence, Net force on central charge will always be zero, irrespective of value of charge q.

Therefore, to find value of q **we have to check equilibrium of any one charge at the corner.**

Step 3 : Force due to all the charges at point C

Force on charge at C **due to** B, $\quad \text{F}_1 = \frac{\text{KQ}^2}{\text{a}^2}$

Force on charge at C **due to** D, $\quad \text{F}_2 = \frac{\text{KQ}^2}{\text{a}^2}$

Force on charge at C **due to** A, $\quad \text{F}_4 = \frac{\text{KQ}^2}{\text{AC}^2} = \frac{\text{KQ}^2}{2\text{a}^2}$

Force on charge at C **due to** q **at centre,** $\quad \text{F}_3 = \frac{\text{KqQ}}{\text{OC}^2} = \frac{2\text{KqQ}}{\text{a}^2}$

Step 4: Apply the equilibrium condition at C:

For the system to be in equilibrium, net force acting on charge at C must be zero.

So, $\vec{F}_3 + \vec{F}_1 + \vec{F}_2 + \vec{F}_4 = 0$

Resultant of F_1 **and** F_2 **(Along** OC**, by symmetry)** $= \sqrt{\text{F}_1^2 + \text{F}_2^2} = \sqrt{2}\,\text{F}_1$

Since $(|\text{F}_1| = |\text{F}_2|)$

Also, F_3**&** F_4 **are along** OC**, Therefore, magnitudes of sum of these forces should be zero.**

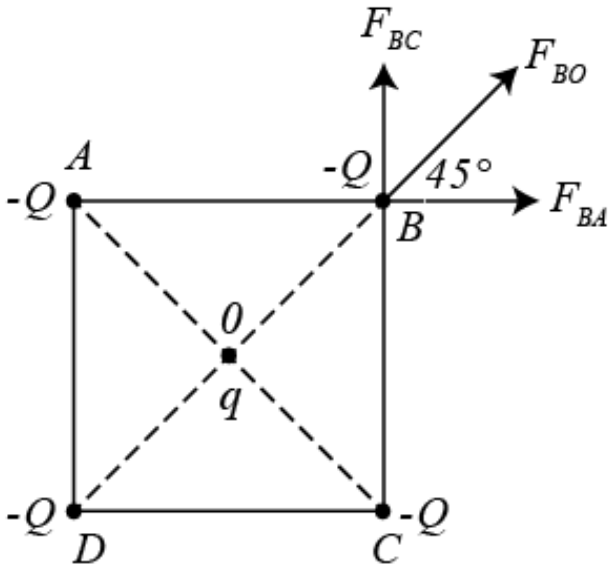
$\Rightarrow \quad |\text{F}_3| + \sqrt{2}\,|\text{F}_1| + |\text{F}_4| = 0$

$\Rightarrow \frac{2\text{KqQ}}{\text{a}^2} + \sqrt{2}\frac{\text{KQ}^2}{\text{a}^2} + \frac{\text{KQ}^2}{2\text{a}^2} = 0$

$\Rightarrow 2\text{q} = -\left(\sqrt{2}\text{Q} + \frac{\text{Q}}{2}\right)$

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$\Rightarrow q = -\frac{Q}{4}(1+2\sqrt{2})$



Q.16 Answer:

0.25

Solution:

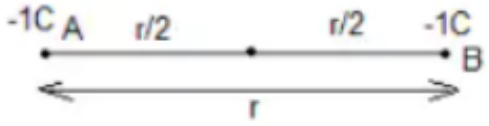
Now total force acting on (-1) at A is

$F_{\text{net}} = \frac{k(-1)(-1)}{r^2} + \frac{k(-1)(q)}{\left(\frac{r}{2}\right)^2} = 0$

$\frac{1}{r^2} = \frac{4q}{r^2}$

$q = \frac{1}{4} = 0.25C$

$\therefore$ **Value of q for these three charges to be in equilibrium** $= 0.25\text{C}$



Q.17 Answer:

$1:3$

Solution:

$T_{AB} = \frac{kq\cdot 2q}{d^2} + \frac{kq\cdot 4q}{(2d)^2} = \frac{3kq^2}{d}$

$T_{BC} = \frac{k\cdot 4q\cdot 2q}{d^2} + \frac{kq\cdot 4q}{(2d)^2} = \frac{9kq^2}{d^2}$

Q.18 Answer:

6×10^{18}

Solution:

Number of protons

$= \frac{1\text{C}}{1.66\times 10^{-19}\text{C}} \approx 6 \times 10^{18}$

Q.19 Answer:

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decreases

Solution:

On touching the metal knob with a positively charged rod some electrons from the gold leaves get transferred to the rod making gold leaves positively charged and they get separated. When a negatively charged rod is touched with metal knob some negative charge flows to the gold leaves lessening the positive charge there and the separation between the leaves decreases.

Q.20 Answer:

$8 \quad \mu C$

Solution:

$8 \quad \mu C$

Final charge $= \frac{2}{1+2} \times \left(12 + 0\right)$

on bigger sphere $= \frac{2}{3} \times 12 = 8 \quad \mu C$

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